Explainable AI – Report

Explaining Anomalies in Credit Card Fraud Detection

Course: Explainable AI

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1. Introduction

Machine learning models are widely used in important real-world tasks such as fraud detection. These models can achieve high accuracy, but they are often difficult to interpret. This creates a problem because users and stakeholders need to understand why a model makes a certain prediction, especially in sensitive applications like financial transactions.

Explainable AI (XAI) aims to make machine learning models more transparent by providing explanations for their predictions. Instead of treating the model as a "black box", XAI methods help us understand how input features influence the output.

In this project, we use two popular XAI methods: SHAP and counterfactual explanations. The goal is to analyze how a fraud detection model makes decisions and to understand the difference between correct predictions and errors. This allows us to better evaluate the behavior of the model beyond standard performance metrics.

2. Research Question

The main research question of this project is:

How can SHAP and counterfactual explanations help explain the difference between true positive and false positive predictions in a fraud detection model?

More specifically, this project aims to understand:

- which features influence predictions the most
- why the model makes mistakes
- how predictions can be changed

3. Dataset and Model

The dataset used in this project is a credit card fraud detection dataset. It contains information about financial transactions and a binary target variable, where 1 represents fraud and 0 represents a normal transaction.

One important characteristic of this dataset is that it is highly imbalanced. Only a very small percentage of transactions are fraudulent, while the majority are normal. This makes the classification task more difficult and requires careful evaluation.

The dataset includes anonymized features labeled V1 to V28. These features are the result of a transformation (such as Principal Component Analysis, PCA), which means their original meaning is not directly interpretable. In addition, the dataset contains Time and Amount features.

A Random Forest classifier is trained on the data. This type of model is powerful and can capture complex relationships between features, but it is not easily interpretable.

The model achieves a ROC-AUC score of 0.96, indicating strong performance in distinguishing between fraud and non-fraud cases. However, due to class imbalance, additional metrics such as precision and recall are required for a more complete evaluation.

4. Methods

4.1 SHAP (SHapley Additive exPlanations)

SHAP is a method used to explain predictions by assigning each feature a contribution value. These values indicate how much each feature increases or decreases the prediction.

The idea of SHAP is based on concepts from game theory. Each feature is treated as a "player" contributing to the final prediction. The SHAP value represents the contribution of that feature.

SHAP can be used for:

- Global explanations, showing which features are important overall
- Local explanations, showing how features influence a single prediction

For example, a positive SHAP value means that a feature increases the probability of fraud, while a negative value decreases it.

4.2 Counterfactual Explanations (DiCE)

Counterfactual explanations provide a different perspective. Instead of explaining why a prediction was made, they answer the question:

"What needs to change in order to get a different prediction?"

For example, if a transaction is classified as fraud, a counterfactual explanation shows how the feature values would need to change for it to be classified as non-fraud.

In this project, we use the DiCE library to generate counterfactual examples. These examples modify only a small number of features while keeping the instance as realistic as possible.

4.3 Comparison of Methods

SHAP and counterfactual explanations complement each other:

- SHAP explains why the model made a prediction
- Counterfactuals explain how to change the prediction

Using both methods together provides a deeper understanding of the model behavior.

5. Evaluation Criteria

The explanations are evaluated based on the following criteria:

1. Interpretability: how easy it is to understand the explanation
2. Usefulness: whether the explanation helps understand model behavior
3. Actionability: whether the explanation suggests meaningful changes

These criteria help assess whether the explanations are practical and informative.

6. Results

6.1 Model Performance

The model achieves a ROC-AUC score of 0.96 (see Figure 1), which indicates strong overall performance.

```
ROC-AUC: 0.9606396386844043

Selected threshold: 0.5

Classification report at selected threshold:
              precision    recall  f1-score   support

           0     0.9995    1.0000    0.9997     71079
           1     0.9674    0.7236    0.8279       123

    accuracy                         0.9995     71202
   macro avg     0.9835    0.8618    0.9138     71202
weighted avg     0.9995    0.9995    0.9994     71202

Confusion matrix:
[[71076     3]
 [   34    89]]
```

Figure 1: Classification report and confusion matrix.

The classification report shows very high performance for the majority class (non-fraud), with precision and recall close to 1.0.

For the fraud class, the precision is 0.9674, while recall is lower at 0.7236. This means that although most predicted fraud cases are correct, the model misses some actual fraud transactions.

The confusion matrix shows that only 3 normal transactions were incorrectly classified as fraud (false positives), while 34 fraud cases were missed (false negatives). This reflects the difficulty of detecting rare events in an imbalanced dataset.

6.2 Global SHAP Analysis

The SHAP bar plot (see Figure 2) shows that the most important features are V12, V14, V4, and V10, with average SHAP values of approximately 0.07, 0.07, 0.06, and 0.05 respectively. These features have the strongest overall impact on the model's predictions.

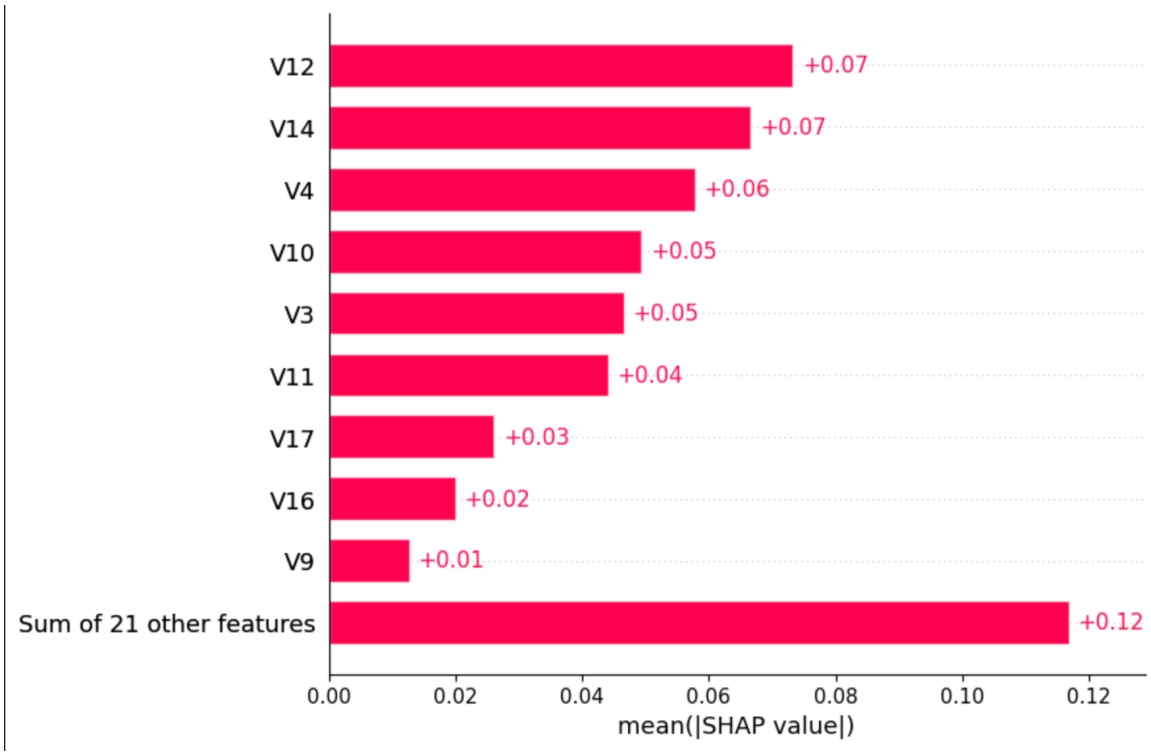

Figure 2: Global SHAP feature importance (bar plot).

The beeswarm plot (see Figure 3) provides more detailed insight into how feature values affect predictions. For example, low values of V12 and V14 are strongly associated with higher fraud probability, while other features show more mixed effects depending on their values.

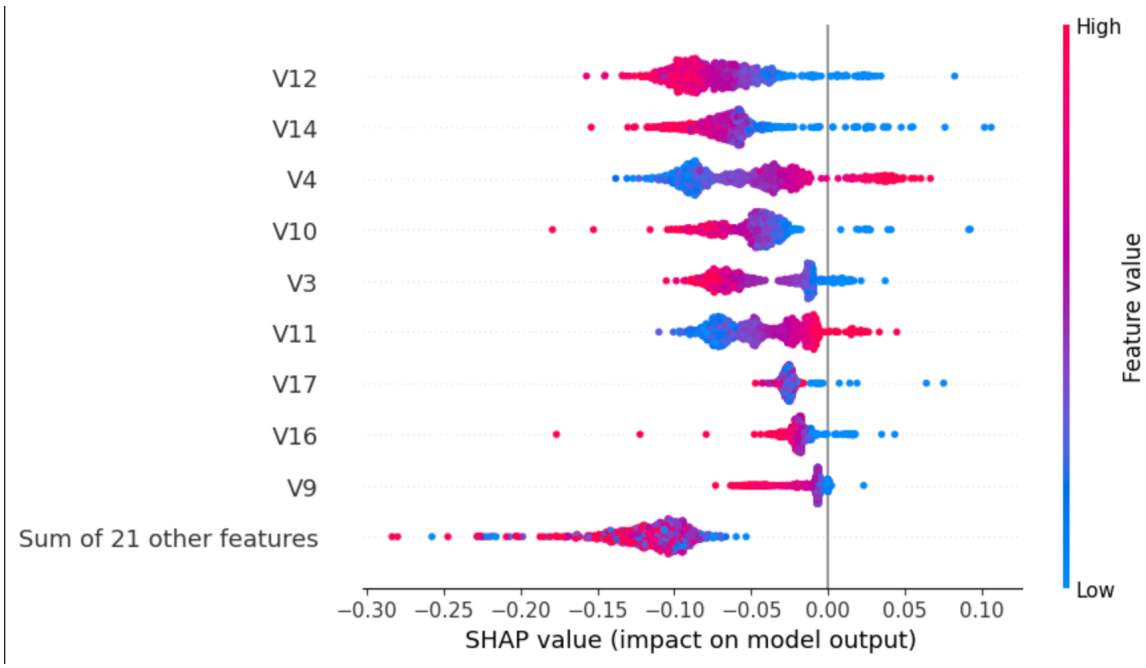

Figure 3: SHAP beeswarm plot showing feature impact distribution.

6.3 Local SHAP Analysis (Individual Cases)

In the true positive case (see Figure 4), the model assigns a fraud probability of 1.0. The most important contributing features are V14 (+0.10), V10 (+0.09), and V12 (+0.09), which strongly increase the prediction towards fraud.

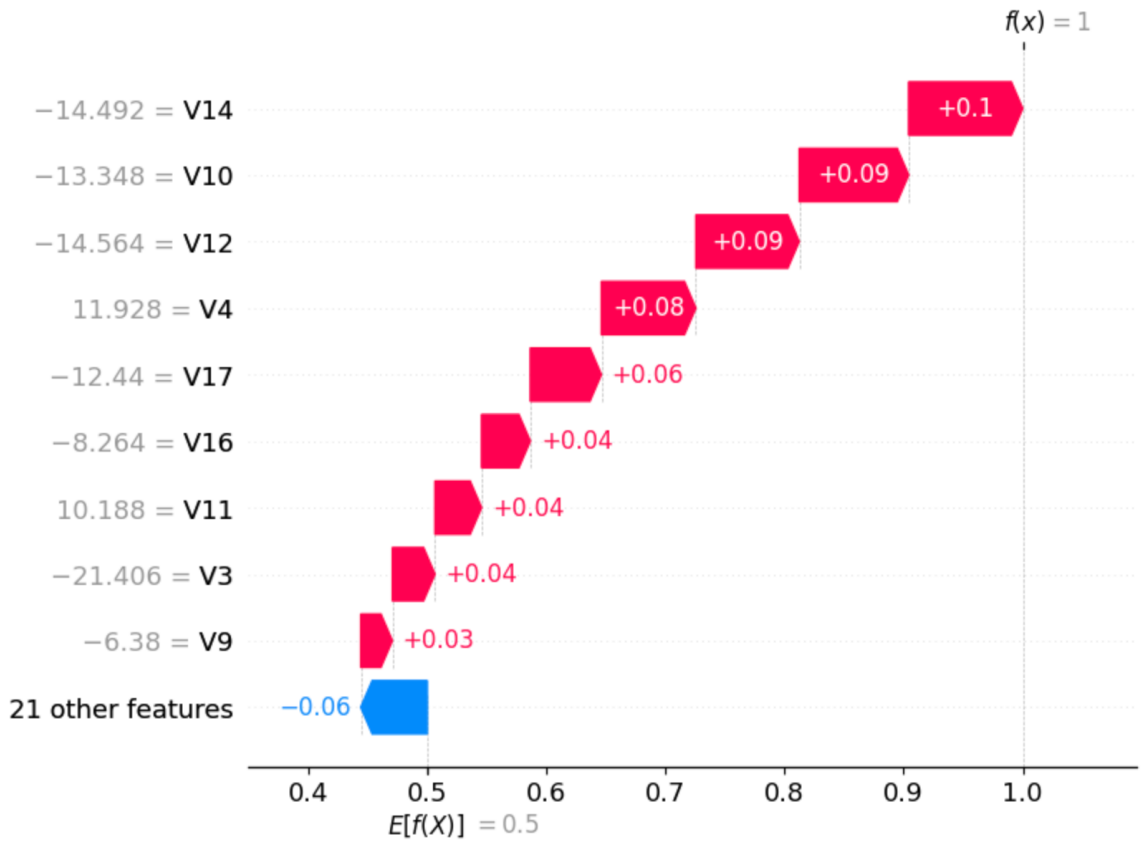

Figure 4: SHAP waterfall plot for the true positive case.

In the false positive case (see Figure 5), the model predicts fraud with a probability of 0.97, even though the transaction is actually normal. Similar features such as V14 (+0.10), V10 (+0.08), and V12 (+0.08) contribute to this prediction.

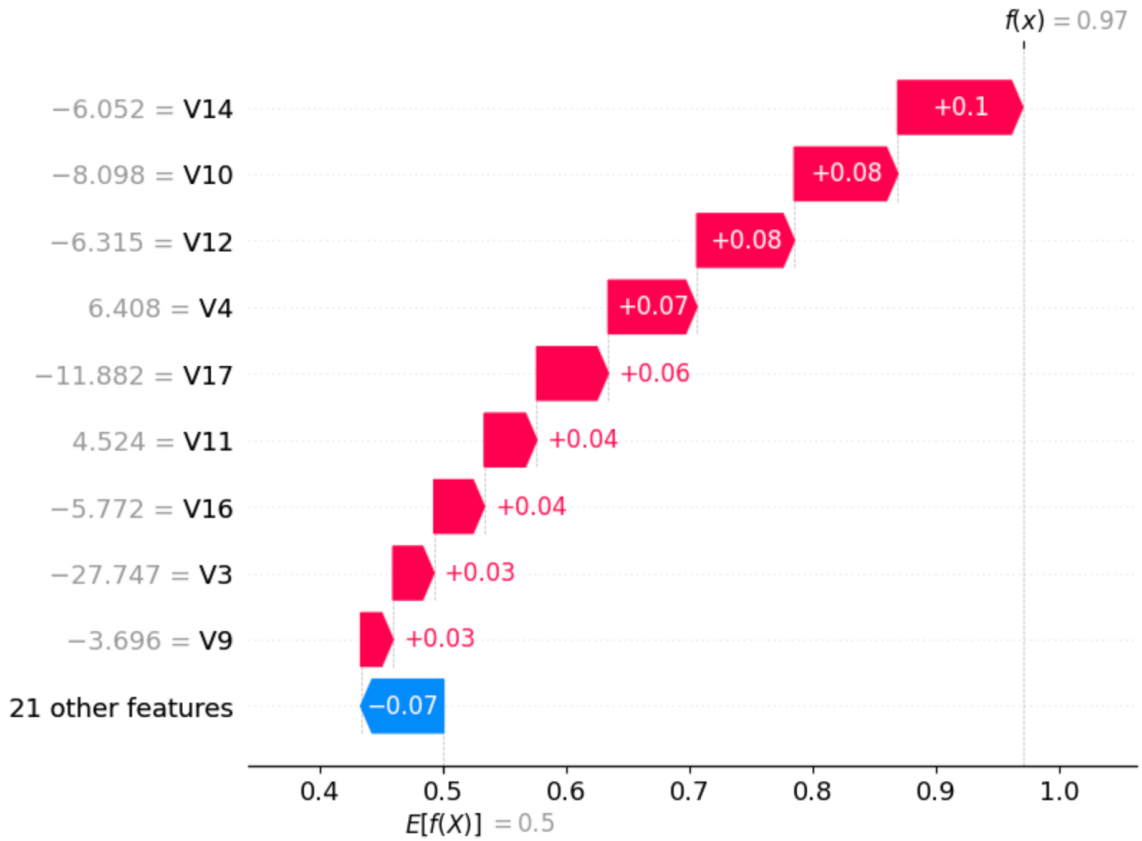

Figure 5: SHAP waterfall plot for the false positive case.

This shows that the model relies on similar feature patterns for both correct and incorrect predictions, which explains why false positives occur.

6.4 Counterfactual Explanations

Counterfactual examples (see Figure 6) show that only the most relevant feature changes are needed to flip the prediction.

```
FALSE POSITIVE original fraud probability: 0.97
TRUE POSITIVE original fraud probability: 1.0

FALSE POSITIVE - changed features only:
```

	CF	Feature	Original	Counterfactual
0	CF 1	V16	-5.772456	7.750407
1	CF 2	V16	-5.772456	10.791772
2	CF 2	V19	0.809988	2.688572
3	CF 3	V8	17.573712	-25.692736
4	CF 3	V11	4.524322	-1.131869

```
TRUE POSITIVE - changed features only:
```

	CF	Feature	Original	Counterfactual
0	CF 1	V7	-15.471612	92.903740
1	CF 1	V9	-6.380125	10.379826
3	CF 2	V6	-2.202710	-22.611503
2	CF 2	V4	11.927512	-5.194707
4	CF 3	V7	-15.471612	114.078778
5	CF 3	V13	0.593759	3.522336

Figure 6: Counterfactual explanations for both cases.

For the false positive case, features such as V16, V19, V8, and V11 need to change significantly to flip the prediction. For example, V16 changes from -5.77 to values above 7.7, indicating a strong influence on the model output.

For the true positive case, features such as V7, V9, V6, and V4 are modified. In particular, V7 changes dramatically (e.g., from -15.47 to over 90), which shows that this feature has a very strong impact on the prediction.

These results confirm that only a small number of features are responsible for changing the model's decision, and that the model is highly sensitive to these variables. However, some counterfactual changes involve large feature shifts, which may not always correspond to realistic transactions.

7. Discussion

The results show that false positives are not random mistakes. Instead, they occur because some normal transactions have feature values similar to fraudulent ones.

SHAP explanations reveal that the same features drive both true positive and false positive predictions. This suggests that the model relies heavily on a small set of features.

Counterfactual explanations show that small changes in these features can change the prediction. This means the model is sensitive to these variables.

By combining both methods, we gain a better understanding:

- SHAP explains which features influence the prediction
- Counterfactuals show how the decision boundary works

This combination helps identify why errors occur and how robust the model is. The results also show that the model assigns very high fraud probabilities (0.97–1.0) even for some incorrect predictions. This suggests that the model is overconfident in certain regions of the feature space. This highlights the importance of combining performance evaluation with explainability methods when deploying models in real-world applications.

8. Limitations

There are several limitations in this analysis:

- The features are anonymized, which makes interpretation difficult
- Counterfactual explanations may not always represent realistic scenarios
- The dataset is highly imbalanced, which can bias the model

These factors limit how much we can interpret the results in real-world terms.

9. Conclusion

In this project, SHAP and counterfactual explanations were used to analyze a fraud detection model. These methods made it possible to better understand how the model makes decisions, rather than only evaluating its performance using standard metrics.

The results show that SHAP provides clear insights into which features are most important and how they influence individual predictions. It helps explain both global patterns in the data and local decisions for specific transactions. At the same time, counterfactual explanations show how small changes in feature values can lead to a different prediction. This makes it easier to understand the decision boundary of the model.

By combining both approaches, we gain a more complete understanding of the model behavior. SHAP explains why a prediction was made, while counterfactual explanations show how the prediction could be changed. This is especially useful for analyzing both correct predictions and errors, such as false positives.

Overall, these methods improve transparency and provide meaningful insights into the model, which is important for building trust in real-world applications like fraud detection. Future work could explore more realistic counterfactual constraints and alternative models to improve robustness.

10. Code repository:

https://github.com/oleksiizadorozhnyi/xai-fraud-detection

11. References

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